

Linear Algebra

Introduction to Matrices

MPS Department|FEU Institute of Technology



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OBJECTIVES

At the end of the discussion, the learner should be able to:

- Define matrix and other related terms.
- Determine the size of a matrix
- Determine the value of an element given its position
- Identify types of matrices



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Definition of a Matrix

a rectangular array of variables, constants or objects in general, in horizontal rows and vertical columns enclosed in brackets.

Matrix

Consider the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

element in i th row, j th column

m rows

n columns

$m \times n$ matrix

Also written as $A = [a_{ij}]$



Matrix: Notation/Symbol

You can represent matrix by capital letter.

Example: **A,B,C,.... X,Y,Z**

$$A=[a_{ij}]$$

where : i = row position
 j = column position



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Order or size of a Matrix

The size of a matrix is described by the notation $m \times n$ (read as m by n).

where: m = number of rows

n = number of columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & a_{ij} & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{jk}]$$



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Example

Suppose $A = \begin{bmatrix} 1 & -3 & 2 & 10 & 0 \\ -2 & 4 & 5 & 1 & 2 \\ -4 & 25 & 6 & 8 & 7 \end{bmatrix}$.

1. How many rows in A are there?
2. How many columns in A are there?
3. What is the size of A ?



Answer:

Suppose $A = \begin{bmatrix} 1 & -3 & 2 & 10 & 0 \\ -2 & 4 & 5 & 1 & 2 \\ -4 & 25 & 6 & 8 & 7 \end{bmatrix}$.

1. How many rows in A are there? **Ans. 3**
2. How many columns in A are there? **Ans. 5**
3. What is the size of A ? **Ans. 3×5 .**

Example:

Consider matrices $A = \begin{bmatrix} -1 & 4 \\ 3 & 6 \\ 0 & -9 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 0 & 4 & -7 \\ 3 & -1 & 5 & 2 \\ 6 & 9 & -3 & 1 \end{bmatrix}$ and

$C = \begin{bmatrix} 1 \\ 5 \\ -10 \end{bmatrix}$. Find their dimensions.



Answer:

- Matrix A has order 3×2 , matrix B is a 3×4 , and matrix C is a 3×1 matrix



Matrix

In the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix}$$

- numbers a_{ij} are called *elements or entries*.
- First subscript indicates the row;
- Second subscript indicates the column.
- The matrix consists of mn elements



Matrix

Example:

Consider matrix $A = \begin{bmatrix} -1 & 0 \\ 3 & 5 \\ -2 & 7 \end{bmatrix}$. The element in $a_{1,2}$ corresponds to the entry found in first row, second column. Thus its value is 0. Likewise, the $a_{2,2} = 5$, $a_{2,1} = 3$, and $a_{3,2} = 7$.



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Example

Suppose $A = \begin{bmatrix} 1 & -3 & 2 & 10 & 0 \\ -2 & 4 & 5 & 1 & 2 \\ -4 & 25 & 6 & 8 & 7 \end{bmatrix}$. Find the value of the following entries:

1. a_{12}
2. a_{32}
3. a_{43}
4. a_{33}
5. a_{31}



Answer:

Suppose $A = \begin{bmatrix} 1 & -3 & 2 & 10 & 0 \\ -2 & 4 & 5 & 1 & 2 \\ -4 & 25 & 6 & 8 & 7 \end{bmatrix}$. Find the value of the following entries:

1. $a_{12} = -3$
2. $a_{32} = 25$
3. $a_{23} = 5$
4. $a_{35} = 7$
5. $a_{31} = -4$



TYPES OF MATRICES

1. Rectangular matrix

Contains more than one element and number of rows is not equal to the number of columns. It has size $m \times n$ where $m \neq n$.

$$\begin{bmatrix} 1 & 1 \\ 3 & 7 \\ 7 & -7 \\ 7 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 0 & 3 & 3 & 0 \end{bmatrix}$$

$$m \neq n$$



TYPES OF MATRICES

2. Column Matrix or Vector:

The number of rows may be any integers, but the number of columns is always 1. Thus, the size should be $m \times 1$, for any positive integer m .

$$\begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$$



TYPES OF MATRICES

3. Row matrix or vector

Any number of columns but with only one row.
Thus, the size should be $1 \times n$, for any positive integer n .

$$\begin{bmatrix} 1 & 1 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 5 & 2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \end{bmatrix}$$



TYPES OF MATRICES

4. Square matrix

The number of rows is equal to the number of columns (a square matrix A has an order of n).

$$\begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix}^{n \times n} \quad \begin{bmatrix} 1 & 1 & 1 \\ 9 & 9 & 0 \\ 6 & 6 & 1 \end{bmatrix}$$

The principal or main diagonal of a square matrix is composed of all elements a_{ij} for which $i = j$.



TYPES OF MATRICES

5. Diagonal matrix

A square matrix where all the elements are zero except those on the main diagonal.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i \neq j$ and

There exists $a_{ij} \neq 0$ where $i = j$



TYPES OF MATRICES

6. Scalar matrix

A diagonal matrix whose main diagonal elements are equal to the same number. Hence, it can be defined as a single number or constant.

$$\begin{bmatrix} a_{ij} & 0 & 0 \\ 0 & a_{ij} & 0 \\ 0 & 0 & a_{ij} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i \neq j$ and $a_{ij} = a$ for all $i = j$ for some constant number a .



TYPES OF MATRICES

7. Unit or Identity matrix (I_n)

A diagonal matrix with 1s on the main diagonal.

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_n = \begin{bmatrix} a_{11} & \cdots & a_{n1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{nn} \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i \neq j$

$a_{ij} = 1$ for all $i = j$



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TYPES OF MATRICES

8. Triangular matrix

A square matrix whose elements above or below the main diagonal are all 0s.

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 8 & 9 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$



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TYPES OF MATRICES

Two Types of Triangular matrix

A triangular matrix may classify as **lower triangular matrix** or **upper triangular matrix**.



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TYPES OF MATRICES

Upper Triangular Matrix

A square matrix whose elements below the main diagonal are all 0s.

$$\begin{bmatrix} a_{ij} & a_{ij} & a_{ij} \\ 0 & a_{ij} & a_{ij} \\ 0 & 0 & a_{ij} \end{bmatrix} \quad \begin{bmatrix} 1 & 8 & 7 \\ 0 & 1 & 8 \\ 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 7 & 4 & 4 \\ 0 & 1 & 7 & 4 \\ 0 & 0 & 7 & 8 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i > j$.

TYPES OF MATRICES

Lower Triangular Matrix

A square matrix whose elements above the main diagonal are all 0s.

$$\begin{bmatrix} a_{ij} & 0 & 0 \\ a_{ij} & a_{ij} & 0 \\ a_{ij} & a_{ij} & a_{ij} \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i < j$



TYPES OF MATRICES

9. Null (zero) matrix - 0

All elements in the matrix are zeros.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$a_{ij} = 0 \quad \text{For all } i, j$$



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Answer:

Identify what specific type of rectangular matrix is given below. Also, determine its dimensions.

1. $A = [-1 \quad 3]$

Answer: a 1×2 row matrix

2. $B = \begin{bmatrix} -1 & 0 \\ 3 & 5 \\ -2 & 7 \end{bmatrix}$

Answer: a 3×2 simply rectangular matrix

3. $C = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Answer: a 4×5 zero matrix

4. $D = \begin{bmatrix} 0 & -7 & 4 \\ 6 & 0 & 5 \\ 3 & -1 & -8 \\ 1 & -2 & 9 \end{bmatrix}$ Answer: a 4×3 matrix

5. $F = [2 \quad 0 \quad 5]$

Answer: a 1×3 row matrix

6. $E = \begin{bmatrix} -1 \\ 3 \\ 1 \\ 4 \\ 0 \end{bmatrix}$ Answer: a 5×1 column matrix



Example:

Identify what specific type of square matrix is given below.

1. $\begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

7. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

2. $\begin{bmatrix} 2 & 1 & 0 \\ 4 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

8. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 6 & 0 & 0 \end{bmatrix}$

3. $\begin{bmatrix} 1 & -2 & 3 & 5 \\ 0 & 0 & 4 & -8 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 5 \end{bmatrix}$

6. $\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

9. $\begin{bmatrix} 2 & -2 & 3 & 5 \\ 1 & 5 & 4 & -8 \\ 0 & 7 & 1 & 10 \\ 0 & 0 & 1 & 5 \end{bmatrix}$



Answer:

1. 2×2 square matrix
2. 3×3 square matrix
3. 4×4 upper triangular Matrix
4. 2×2 Diagonal Matrix
5. 3×3 Identity matrix
6. 3×3 scalar matrix
7. 2×2 zero matrix
8. 3×3 lower triangular Matrix
9. 3×3 square matrix



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